

7(a)(i) When the wavelength of the wave is comparable to slit width, diffraction is most significant.

(ii) Low-pitched notes have longer wavelengths that's more comparable to the width of door gap and undergo more diffraction through the door.

(b)(i) Distance D must be much longer than distance two adjacent slits of the grating

(ii) Slit separation, $d = 1/p$

$$\text{path difference } x = d \sin \theta = 1/p \sin \theta$$

(iii) Hence Constructive interference occurs when

$$1/p \sin \theta = n \lambda \text{ where } n = 0, 1, 2, \dots$$

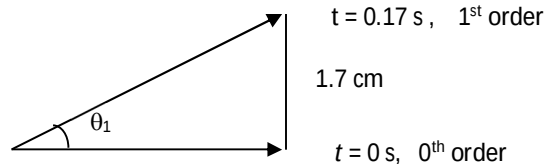
Where n is the order of the spectrum and θ is the angle of diffraction between the diffracted ray and the direction of incident light.

c (i) angular speed of grating, $\omega = 2 \pi / T = 2 \pi / 3 = 2.09 = 2.1 \text{ rad.s}^{-1}$

(ii) Peak C corresponds to the zeroth order of the interference pattern whereas peaks B and D the first order and Peaks A and E the second order.
The height of each peak is due to diffraction effect.

(iii)

Hence, time to shift from zeroth order to the first order through 1.7 cm on the screen is $1.7 \times 0.1 = 0.17\text{s}$



$$\theta_1 = (2\pi / 3) 0.17 \text{ rad} = 0.355 \text{ rad} = 0.36 \text{ rad}$$

Similarly time from the zeroth order to 2nd order = $3.7 \times 0.1 = 0.37\text{ s}$

$$\theta = (2\pi / 3) 0.37 \text{ rad} = 0.773 \text{ rad} = 0.77 \text{ rad}.$$

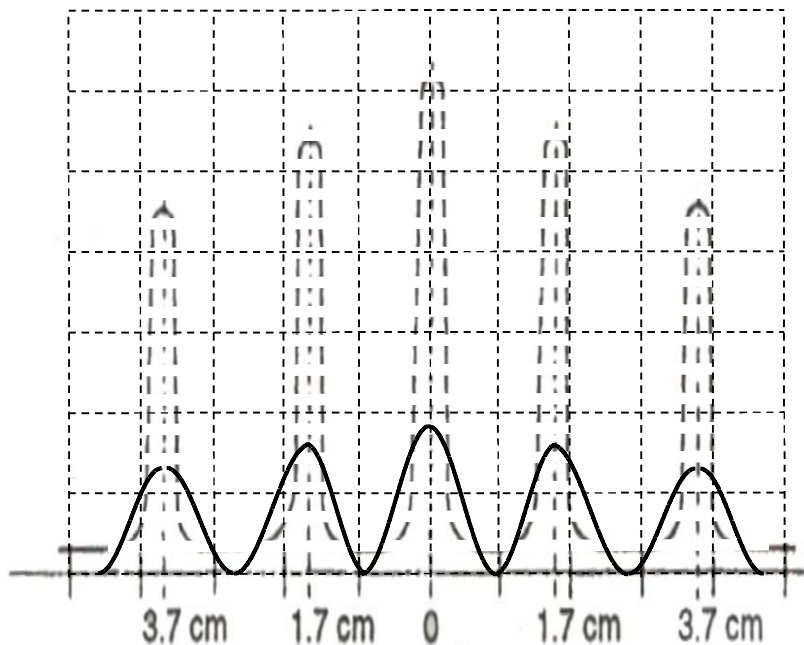
(iv) Using the grating equation

$$\sin \theta_2 = 2 \times 5.5 \times 10^5 \times \lambda$$

$$\lambda = 636 \text{ nm}$$

(v) Peak E is preferred as the angle θ is larger and the percentage error for calculating the wavelength is smaller.

(vi) Since the slit separation remain the same, the fringe separation remains the same. However, the peaks are less intense (poorer contrast) and less sharp (broader and less defined).



8 (a) (i) most alphas have small deflection